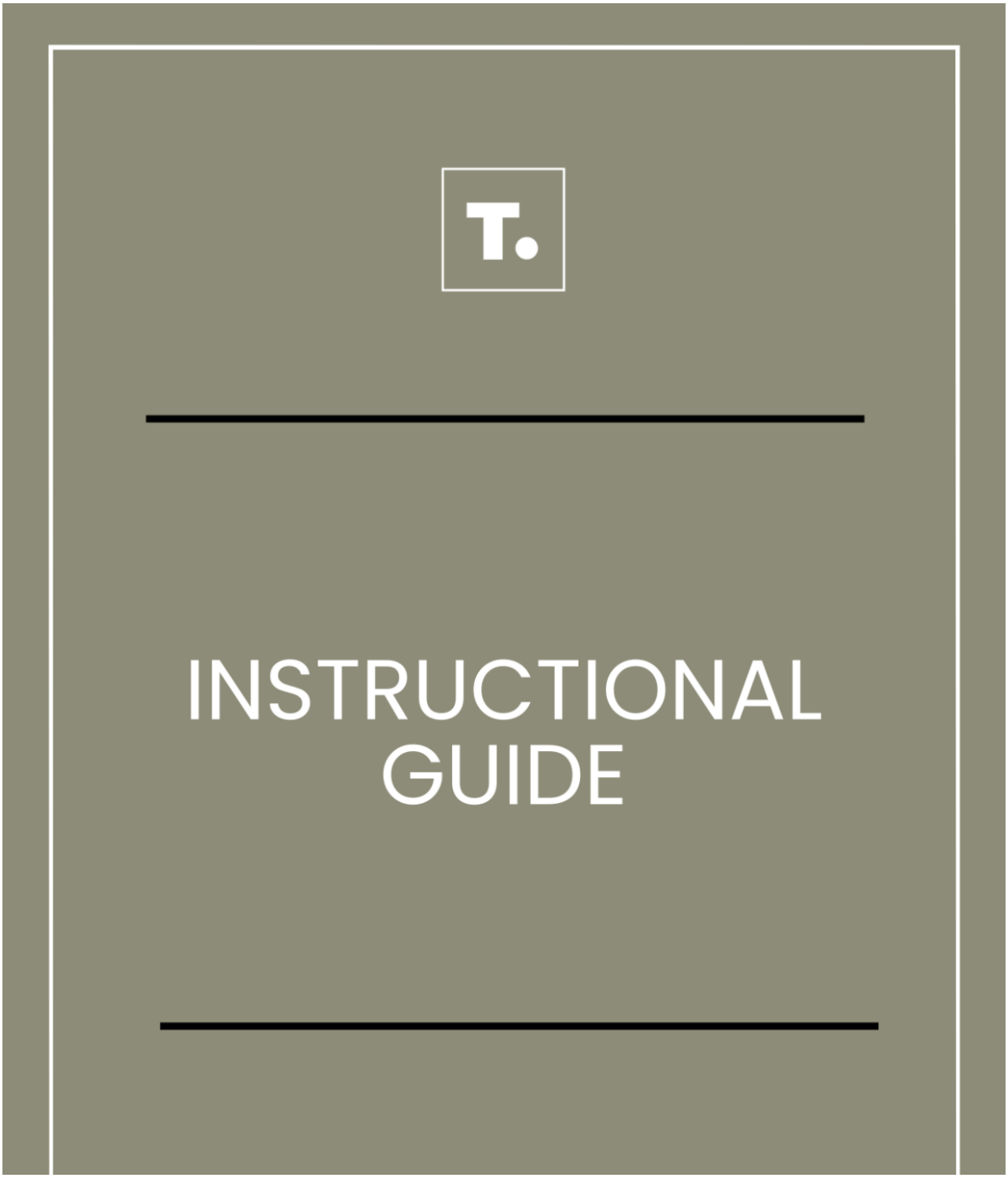
Phishing and Ransomware Scenario Simulation Lab

INSTRUCTOR GUIDE



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1. Purpose of the Lab

This lab is designed to provide students with practical exposure to a phishing-based attack scenario that escalates into ransomware-like behavior within a controlled virtual environment. The lab focuses on attack execution, detection, logging, and incident analysis rather than real malware deployment. The exercise supports learning outcomes related to cybersecurity operations, incident response, and digital forensics.

2. Learning Outcomes

Upon successful completion of this lab, students should be able to:

- Explain phishing as a common initial attack vector
- Configure and run a controlled phishing campaign
- Safely simulate ransomware behavior using benign scripts
- Identify and analyze host-based and network-based logs
- Reconstruct an incident timeline using collected evidence
- Document findings in a structured incident report

3. Lab Prerequisites

3.1 Student Knowledge Requirements

- Basic understanding of operating systems (Windows & Linux)
- Introductory knowledge of networking concepts
- Familiarity with cybersecurity terminology

3.2 Infrastructure Requirements

- VirtualBox or equivalent hypervisor
- Minimum host specifications:
  - ✓ GB RAM (lab designed to operate under constrained resources)
  - ✓ Limited storage (logical evidence acquisition acceptable)

4. Lab Environment Overview

4.1 Virtual Machines Required

VM Name	Operating System	Role
Victim	Windows 10 Pro	End user workstation
Utility	Debian 12	Mail server, IDS, logging
Attacker	Kali linux	Phishing platform

4.2 Network Design

- Host-only networking for internal communication
- NAT enabled temporarily for package installation
- No Domain Controller or File Server required

➢ The absence of a Domain Controller is intentional due to hardware constraints.

5. Tools Used

- GoPhish (phishing framework)
- Postfix & Dovecot (mail services)
- Suricata (IDS)
- PowerShell (benign ransomware simulation)
- tcpdump & Wireshark (network analysis)
- Windows Event Viewer (host-based logging)

6. Lab Execution Summary

Phase 1: Phishing Setup

Students configure GoPhish to send a phishing email to a single test account. Email delivery is validated through mail server logs and the recipient's mailbox.

Phase 2: Payload Simulation

Students execute a provided benign PowerShell script that simulates ransomware behavior by locking files in specific directories and creating a ransom note.

Phase 3: Detection and Logging

Students examine:

- Windows Security and PowerShell logs
- Mail server logs (Postfix, Dovecot)
- Network traffic captured via tcpdump
- IDS alerts (Suricata)

Phase 4: Analysis

Students reconstruct the incident timeline and document findings in an incident report.

7. Expected Student Outputs

Students should submit:

- Completed student workbook
- Incident report
- Screenshots of key evidence
- Timeline reconstruction

Optional:

- PCAP files
- Exported log files

8. Assessment Guidance

Key areas to assess:

- Correct setup of the lab environment
- Successful phishing email delivery
- Proper execution of the benign simulation
- Evidence of log collection and analysis

- Quality and clarity of the incident report
- Reflection on limitations and lessons learned

Perfect tool output is not required. Students should be assessed on understanding and analysis rather than flawless logging.

9. Common Issues and Notes

- Sysmon may produce limited event data in virtualized environments
- IDS alerts may not trigger for benign payloads
- Students may rely on screenshots and logical evidence acquisition due to storage constraints

These limitations should be acknowledged, not penalized.

10. Ethical and Safety Considerations

- No real malware is used
- All activities occur in isolated virtual machines
- The PowerShell script is benign and reversible
- Students must not deploy tools outside the lab environment

11. Conclusion

This laboratory exercise provides a structured and practical approach to teaching phishing-based attack scenarios and basic incident response techniques within a controlled and ethical environment. By guiding students through the full lifecycle of a simulated cybersecurity incident, the lab reinforces both theoretical understanding and hands-on technical skills that are essential in real-world cybersecurity roles. The use of a benign ransomware simulation ensures that students gain meaningful experience without exposing systems to actual malicious software. From an instructional perspective, this lab allows educators to assess a wide range of competencies, including technical configuration, attack analysis, evidence handling, and professional documentation. Overall, this lab serves as an effective teaching tool for introducing students to phishing attacks, ransomware concepts, and incident response processes in a safe, reproducible, and educationally sound manner.